

Is the square matrix $\begin{pmatrix} 0 & 0 & a \\ 0 & b & 0 \\ c & 0 & 0 \end{pmatrix}$ with nonzero entries along the "diagonal" from top right to bottom left, and zero entries elsewhere considered a diagonal matrix?

Answer:

The short answer is **no**. In standard linear algebra, a "diagonal matrix" specifically refers to entries along the **main diagonal** (top-left to bottom-right).

A square matrix with nonzero entries only along the path from the top-right to the bottom-left is called an **anti-diagonal matrix** (or sometimes a counter-diagonal matrix).

The Mathematical Difference

To be technically precise, we define these two types of matrices based on their indices (i, j) for an $n \times n$ matrix:

- **Diagonal Matrix:** All entries $a_{i,j}$ must be zero unless $i = j$.
- **Anti-Diagonal Matrix:** All entries $a_{i,j}$ must be zero unless $i + j = n + 1$.

Properties

Note that many "nice" properties of diagonal matrices do not carry over to anti-diagonal matrices.

For example, the determinant of a diagonal matrix is always the product of all its diagonal entries. This is not always the case for anti-diagonal matrices. For example,

$$\begin{vmatrix} 0 & 0 & a \\ 0 & b & 0 \\ c & 0 & 0 \end{vmatrix} = -abc \text{ and } \begin{vmatrix} 0 & 0 & 0 & a \\ 0 & 0 & b & 0 \\ 0 & c & 0 & 0 \\ d & 0 & 0 & 0 \end{vmatrix} = abcd.$$

In general, for an $n \times n$ anti-diagonal matrix, the determinant is $(-1)^n \times$ product of all the anti-diagonal entries.

When you multiple two diagonal matrices (of the same size), the product is again a diagonal matrix. When you multiple two anti-diagonal matrices (of the same size), the product is a diagonal matrix, instead of another anti-diagonal matrix.